

# The Mishkin Test as a Return Regression

Teaching note

Sloan (1996), *The Accounting Review* 71 (3): 289–315

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## Sloan's claim rests on a coefficient test

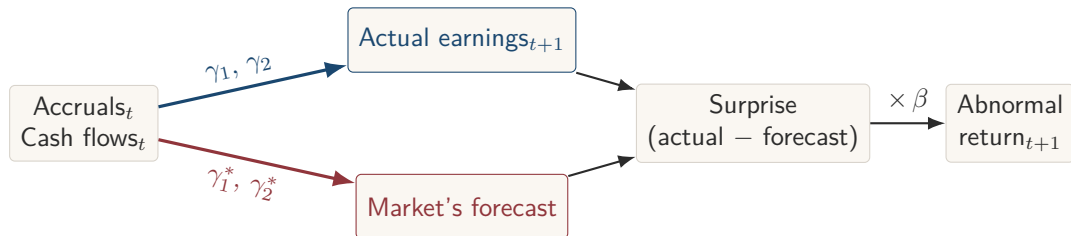
*"However, stock prices are found to act as if investors 'fixate' on earnings, failing to reflect fully information contained in the accrual and cash flow components of current earnings until that information impacts future earnings."*

Sloan (1996, p. 289)

The evidence for "act as if" is the Mishkin (1983) test: Sloan Table 5 (p. 305) reports a likelihood ratio (LR) statistic of 180.91 ( $p = 0.000$ ).

- This note explains what the test estimates, why Sloan chose it, and what later papers found it can and cannot show.

## Efficiency requires equal weights



Forecasting equation: the weights (persistence coefficients) with which the data turn accruals and cash flows into next year's earnings.

Pricing equation: the weights investors appear to have used;  $\beta$ , the earnings response coefficient, is the return per unit of surprise.

- If the market is efficient, the starred weights equal the unstarred weights, and the surprise cannot be predicted at date  $t$ .

## A weight gap implies a predictable return

Consider a firm whose accruals equal 0.10 of assets, using Sloan's Table 5 estimates (p. 305), with cash flows held fixed and  $\beta = 1.894$ .

| Step                         | Calculation                                       | Result  |
|------------------------------|---------------------------------------------------|---------|
| Earnings the data deliver    | $\gamma_1 \times 0.10 = 0.765 \times 0.10$        | 0.0765  |
| Earnings the market expected | $\gamma_1^* \times 0.10 = 0.911 \times 0.10$      | 0.0911  |
| Expected surprise            | $0.0765 - 0.0911$                                 | -0.0146 |
| Expected abnormal return     | $\beta \times (-0.0146) = 1.894 \times (-0.0146)$ | -2.8%   |

- A disappointment that is knowable at date  $t$  is a predictable return; the test is designed to detect this.

## The test compares two fits of one system

Sloan eqs. (10)–(11), p. 302; joint iterative weighted nonlinear least squares, p. 303:

$$Earnings_{t+1} = \gamma_0 + \gamma_1 Accruals_t + \gamma_2 CashFlows_t + v_{t+1} \quad (10)$$

$$AbnRet_{t+1} = \beta (Earnings_{t+1} - \gamma_0 - \gamma_1^* Accruals_t - \gamma_2^* CashFlows_t) + \varepsilon_{t+1} \quad (11)$$

**Step 1:** fit with  $\gamma^*$  free; record the sum of squared residuals,  $SSR^u$ .

**Step 2:** refit forcing  $\gamma_1^* = \gamma_1$  and  $\gamma_2^* = \gamma_2$ ; record  $SSR^c$ .

**Step 3:**  $LR = 2n \log(SSR^c/SSR^u)$ , asymptotically  $\chi^2(q)$  (Sloan eq. (12));  $q = 2$ , 5 percent critical value 5.99.

$v$ : earnings forecast error;  $\varepsilon$ : return noise;  $n = 40,679$  firm-years.

- A large LR indicates that forcing the market's weights to equal the data's weights worsens the fit materially.

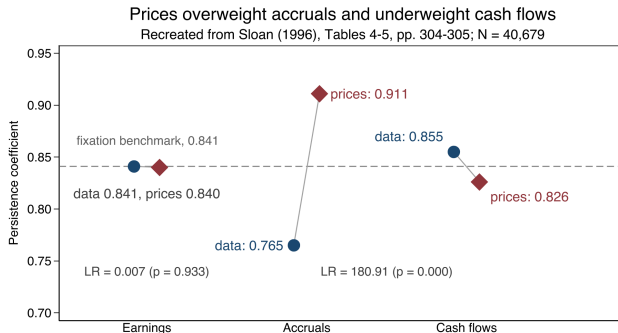
# Prices appear to overweight accruals

Circles: persistence in the data.

Diamonds: persistence implied by prices.

Fixation predicts both diamonds at 0.841, the persistence of total earnings (Sloan Table 4, p. 304).

■ Sloan rejects rational pricing of the components, in the direction of fixation but not at its magnitude (0.841).



## Three features suit the test to Sloan's H2(i)

1. **Magnitudes.** H2(i) “makes predictions about both the direction and the magnitude of deviations” (p. 292); only the starred coefficients can be compared with 0.841.
  2. **Robustness.** The tests “remain valid tests of market efficiency regardless of whether the forecasting equation has omitted variables” (fn. 13, p. 302).
  3. **Joint estimation.** A two-step approach “uses generated regressors in the stock return equation, resulting in inconsistent standard errors” (fn. 14, p. 303), i.e., first-stage fitted values used as data.
- Later papers evaluate each claim: joint estimation (slides 8–10), robustness (11–12), and magnitudes (13).

## Substitution yields a return regression

Substitute Sloan eq. (10) into eq. (11):

$$AbnRet_{t+1} = \beta v_{t+1} + \beta(\gamma_1 - \gamma_1^*) Accruals_t + \beta(\gamma_2 - \gamma_2^*) CashFlows_t + \varepsilon_{t+1}$$

$$\gamma^* = \gamma \iff \text{zero return slopes}$$

$v_{t+1}$  is unforecastable at date  $t$ , so it joins the error

$$\beta(\gamma_1 - \gamma_1^*) = -0.28$$

1.894 (0.765 - 0.911), Sloan Table 5

- Given  $\beta \neq 0$ , the Mishkin restriction holds if and only if accruals and cash flows do not predict abnormal returns (Lewellen 2010, p. 460).

## Unconstrained Mishkin estimates equal OLS

```
* truth: Sloan Table 5 values
gen e1 = 0.01 + 0.765*acc ///
      + 0.855*cf + v
gen ar = 1.894*(e1 - 0.01 ///
      - 0.911*acc - 0.826*cf) + e
mishkin e1 ar acc cf // system
reg ar e1 acc cf // b, d1, d2
* gamma1* = -d1/b; gamma2* = -d2/b
```

|                             | Truth | Mishkin NLS  | From OLS     |
|-----------------------------|-------|--------------|--------------|
| $\gamma_1$ (accruals)       | 0.765 | 0.764        | 0.764        |
| $\gamma_2$ (cash flows)     | 0.855 | 0.849        | 0.849        |
| $\gamma_1^*$                | 0.911 | <b>0.919</b> | <b>0.919</b> |
| $\gamma_2^*$                | 0.826 | <b>0.832</b> | <b>0.832</b> |
| $\beta$                     | 1.894 | 1.896        | 1.896        |
| Test statistic, $\chi^2(2)$ |       | LR = 245.2   | Wald = 245.7 |

Simulated data (seed 1996),  $N = 40,679$ . Sloan eq. (11) implies OLS slopes  $b = \beta$  on  $Earnings_{t+1}$  and  $\delta_k = -\beta\gamma_k^*$  on each component, so  $\gamma_k^* = -\delta_k/b$ .

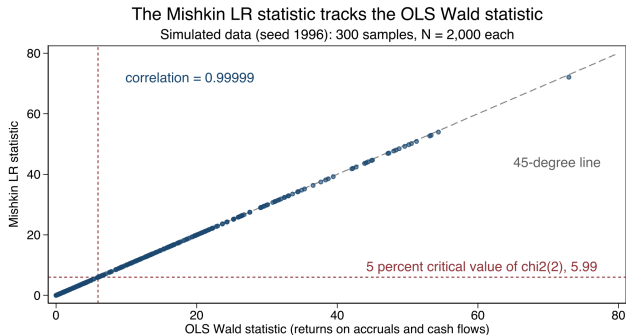
■ With a free pricing intercept, the nonlinear system and OLS agree to within  $10^{-6}$ .

# The LR and OLS Wald tests rarely disagree

Wald statistic: the  $\chi^2$  test that both slopes are zero in the OLS regression of abnormal returns on accruals and cash flows.

Across 300 samples, the two tests disagree at the 5 percent level once.

- In large samples the framework appears to offer no statistical advantage over OLS (KLW 2007, p. 1111).



## Omitted mispriced correlates shift $\gamma_1^*$

Add  $Z_t$  (e.g., growth): data weight  $\gamma_3$ , market weight  $\gamma_3^*$ , and  $Z_t = d \text{Accruals}_t + u_t$ .  
With  $Z_t$  omitted, the estimates (hats) converge in large samples to:

$$\hat{\gamma}_1^* - \hat{\gamma}_1 \rightarrow \underbrace{(\gamma_1^* - \gamma_1)}_{\text{accrual mispricing}} + \underbrace{d(\gamma_3^* - \gamma_3)}_{\text{mispricing of } Z, \text{ inherited}} \quad (\text{KLW eq. (11), p. 1088})$$

$Z$  rationally priced ( $\gamma_3^* = \gamma_3$ )  
inherited term zero; Sloan's fn. 13 holds

$Z$  mispriced ( $\gamma_3^* \neq \gamma_3$ )  
accruals inherit the mispricing of  $Z$

*"Only if the omitted variables are rationally priced is their exclusion irrelevant."*

Kraft, Leone, and Wasley (KLW 2007, p. 1081)

■ The test assigns to accruals the mispricing of any omitted correlate.

## Correctly priced accruals can appear mispriced

```
gen z = 0.5*acc + rnormal(0,0.08)
gen e1 = 0.01 + 0.765*acc ///
      + 0.855*cf + 0*z + v
* market misprices z only; v, e: noise
gen ar = 1.894*(e1 - 0.01 ///
      - 0.765*acc - 0.855*cf ///
      - 0.292*z) + e
mishkin e1 ar acc cf // Z omitted
mishkin e1 ar acc cf z // Z included
```

|                                   | Truth | Z omitted             | Z included            |
|-----------------------------------|-------|-----------------------|-----------------------|
| $\gamma_1$ (accruals, data)       | 0.765 | 0.762                 | 0.762                 |
| $\gamma_1^*$ (accruals, prices)   | 0.765 | <b>0.916</b>          | <b>0.776</b>          |
| $\gamma_2$ (cash flows, data)     | 0.855 | 0.853                 | 0.853                 |
| $\gamma_2^*$ (cash flows, prices) | 0.855 | 0.849                 | 0.851                 |
| $\gamma_3$ (Z, data)              | 0.000 |                       | 0.000                 |
| $\gamma_3^*$ (Z, prices)          | 0.292 |                       | 0.281                 |
| LR statistic                      |       | 215.0 [ $\chi^2(2)$ ] | 917.1 [ $\chi^2(3)$ ] |

Simulated data (seed 1996),  $N = 40,679$ ;  
analytical gap  $0.292 \times 0.5 = 0.146$  (Sloan's Table 5 gap, by construction).

■ With  $Z$  omitted, correctly priced accruals show  $\gamma_1^* = 0.916$ .

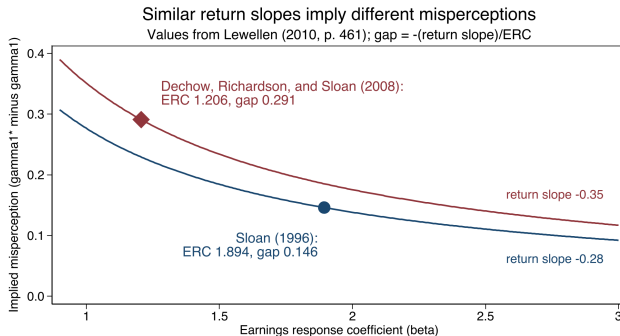
## The implied bias varies inversely with ERC

From slide 8, the return slope on accruals is  $g = \beta(\gamma_1 - \gamma_1^*)$ , so

$$\gamma_1^* - \gamma_1 = -g/\beta$$

The data identify  $g$ . Reading it as an earnings misperception requires dividing by the earnings response coefficient (ERC),  $\beta$ .

- Similar return slopes ( $-0.28$ ,  $-0.35$ ) yield biases of 0.146 and 0.291 because the ERCs differ.



## Pooled LR tests can overstate significance

Simulation with the null true: accruals do not predict returns on average, but the return slope on accruals varies randomly by year (i.e., residuals are correlated across firms within a year).

| Test at the 5 percent level (a correct test rejects 5%)            | Rejection rate |
|--------------------------------------------------------------------|----------------|
| Pooled Mishkin LR, $\chi^2(2)$                                     | <b>35.6%</b>   |
| Pooled OLS Wald, $\chi^2(2)$                                       | <b>35.6%</b>   |
| Fama-MacBeth $t$ (one regression per year; $t$ -test on 30 slopes) | <b>5.8%</b>    |

500 replications (seed 1996); 30 years  $\times$  500 firms. Sloan: LR = 180.91 pools 40,679 firm-years (p. 305); the hedge-return  $t = 4.71$  uses 30 years (p. 307).

- In KLV Table 3 (pp. 1097–1098), the  $t$ -statistic on  $\gamma_1 - \gamma_1^*$  falls from  $-5.41$  pooled to  $-1.68$  year by year.

## Sloan's Table 5 shows predictability only

Sloan Table 5 is, in large samples, a return regression; Tables 6 to 8 document the same predictability with more conservative inference (30 annual observations).

*“However, precise inferences as to whether the magnitudes of the deviations from rational expectations are consistent with a naive fixation on earnings are sensitive to model specification.”*

Sloan (1996, pp. 305–306)

A skeptic would note that the concentration of returns at earnings announcements (Sloan Table 8) speaks to earnings expectations directly.

- The starred coefficients are an interpretation of a return slope, conditional on the pricing model and the included variables.

# Summary

- In large samples, the Mishkin test is a regression of abnormal returns on lagged information: its unconstrained estimates reproduce OLS, and its LR statistic tracks the OLS Wald statistic. The framework offers an interpretive decomposition rather than a statistical advantage.
- The decomposition attributes return predictability to misperceived earnings persistence by assumption. An omitted, mispriced correlate shifts the starred coefficient, and the implied bias varies inversely with the estimated ERC.
- Pooled LR statistics inherit the inference problems of pooled return regressions, which suggests that Sloan's portfolio and Fama-MacBeth evidence carries the more robust inference.

Mishkin (1983) · Abel and Mishkin (1983) · Kraft, Leone, and Wasley (2007) · Lewellen (2010)  
[github.com/sbb151/acctg597a-research-design](https://github.com/sbb151/acctg597a-research-design)